



Natural Language Integration into a React Based User Interface



Cymantha Blackmon, Habiba Morsy, Essence Toone, Mostafa Moamen; *Advisors:* E. Mema, PhD, G. Avirappattu, PhD
The Dorothy and George Hennings College of Science, Mathematics, and Technology, Kean University, 1075 Morris Ave, Union, NJ 07083



INTRODUCTION

In recent years, the advancement of conversational based natural language models has led to an increase in demand for both a personalized and immersive user experience. Applications that integrate large language models (LLMs) are rapidly expanding across the United States, having the potential to revolutionize communication and improve relationships between a company and its customers.

The **goal** of this project is to develop an application capable of receiving an end user's question, responding using data corresponding to **Carnegie Classifications for Institutions of Higher Education** (CCIHE) and updating the database with new question-answer pairs.

CURRENT METHODOLOGY

Customer Input and Storage

- An end user inputs name, email address and question corresponding to CCIHE information. That data is stored in a users database.

Response Generation

- A response is fetched from a model capable of performing Retrieval Augmented Generation (RAG), a method of retrieving information from a particular data source and generating a response tuned to the user's specific inquiry.

Customer Follow Up and Feedback

- Once reviewed, and edits made if needed, by an administrator, the user is emailed a response.

QA Database Update

- The question answer pair are added to a database that can be used to further train the model

RESULTS

Welcome to Carnegie Classifications for Institutions of Higher Education!
By submitting, you consent to the use of your question in the development of our artificial intelligence based interface.

Name

Email Address

Question

Send

Upon entering the details within the form above, the information is stored in a secure database accessible by only administrators responsible with dispatching corresponding responses. Users are notified that their respective inquiry may be used to query a large language model.

GPT 3.5 Fine Tuned
This model has been fine tuned on our Q/A data

Question

In one sentence, describe Kean University.

Retrieve

Kean University is a comprehensive public university that educates a diverse, multicultural student population in New Jersey, including more than 100 undergraduate and 50 graduate degree programs in the liberal arts and sciences, architecture, design, health, and related professional disciplines.

GPT 4 Preview

Question

Describe Kean University in one sentence.

Retrieve

Kean University is a public university located in Union and Hillside, New Jersey, known for offering a wide range of undergraduate and graduate programs, with a strong emphasis on research and global education.

Utilizing the above, administrations are provided access to query an LLM of preference (GPT 3.5, GPT 4, Llama, etc) using the user-submitted question. This process is instrumental towards the continual refinement of our question-and-answer database.

CONCLUSION

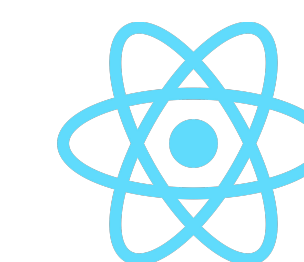
This project exemplifies seamless integration of LLM models into React-based interfaces, enhancing user experiences. Leveraging RAG methodology, it efficiently handles user queries regarding CCIHE. This methodology ensures secure data handling and facilitates effective communication between users and administrators. Through continual feedback and database updates, this project sets a foundation for future advancements in conversational AI and user-centric applications.

Future Directions: As development continues, our goal is to develop a well defined and integrated user interface that can retrieve a question and display a refined response.



*The following QR code can be used to access a **beta** version of the aforementioned user interface*

RESOURCES



React.dev



Vite.js.dev



LangChain



Material UI



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